MILITARY SPECIFICATION

PRESSURE CARTRIDGE ASSEMBLY LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification contains requirements not normally covered by drawings and provides Quality Assurance provisions for the parts, loading assembling, and packing for the Pressure Cartridge Assembly used in the M87 and M88 Canister Assemblies.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-223 - Propellant, Black Powder

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for shipment and storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U S Army Armament, Munitions and Chemical Command, Attn AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-STD-286 - Propellant, Solid, Sampling, Examination and Testing
MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS

9366511 - Pressure Cartridge Assembly

9366635 - Cartridge Case Assembly with Primer

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 173.94 - Hazardous Materials Regulations of the Department of Transportation

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1949 Edition and Revisions) available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite “49 CFR 173.94, (latest revision).”)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Propellant, M1. The propellant shall comply with MIL-STD-652 Propellant Ml, Form C, Type I, with grain diameter of .13 ± .01 inch.

3.1.2 Propellant, black powder . The propellant shall comply with MIL-P-223.

3.2 Pressure cartridge assembly. The cartridge assembly shall comply with all requirements specified on Drawing 9366511 and with all requirements specified therein.

3.3 Moisture content.

3.3.1 Propellant, M1. The moisture content of the M1 propellant used in loading the cartridge assembly shall not exceed 0.50 percent when tested as specified in 4.5.2.

3.3.2 Propellant, black powder . The moisture content of the propellant used in loading the cartridge assembly shall not exceed 0.15 percent when tested as specified in 4.5.3.

3.4 Environmental conditions. The cartridge assembly shall meet the requirements of 3.5 when tested at temperature range of -35°F to +145°F.

3.5 Functioning,.

3.5.1 Impact sensitivity, primer. The primer in the cartridge case assembly when tested in an impact fixture shall fire when impacted with 27 inch ounces of energy, when tested as specified in 4.5.4.1.

3.5.2 Pressure/time values. (+140°F +5°F). The cartridge assembly when tested in a closed bomb test fixture as specified in 4.5.4.2 shall demonstrate the following pressure/time values:

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a.	Maximum Peak Pressure	16K psi
b.	Minimum Pressure at .10 seconds	9.5K psi
c.	Minimum Pressure at .20 seconds	8K psi
d.	Minimum Pressure at .40 seconds	7.2K psi
e.	Maximum Time to Peak Pressure	0.08 seconds

3.5.3 Pressure/time value. (-30°F -5°F). The cartridge assembly when tested in a closed bomb test fixture as specified in 4.5.4.2 shall demonstrate the following pressure/time values:

a.	Minimum Peak Pressure	8K psi
b.	Minimum Pressure at .10 sec	9.5K psi
c.	Minimum Pressure at .20 sec	8K psi
d.	Minimum Pressure at .40 sec	7.2K psi
e.	Maximum time to peak pressure	.05 psi

3.6 First article. This specification makes provision for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated and furnished in a thorough, workmanlike manner. They shall be free of burrs, cracks, chips, sharp edges, unblended radii, surface defects, dirt, grease, hot rolling defects, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First article inspection

b. Quality conformance inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample quantities as indicated:

Part Participation	Drawing	Quantity
Propellant, Ml, Form C, Type I	MIL-STD-652	10 gram
Propellant, Black Powder	MIL-P-223	10 gram
Cartridge Case Assembly	9366635	25
Pressure Cartridge Assembly	9366511	125
Disc	9366605	15
Foam	9366604	15
Paper, Filter	9366607	15
Wadding	9366606	15

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70668(AR)

PARAGRAPH	TITLE: Cartridge Assembly and Components		SHEET 1 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%.	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Black powder (MIL-P-223) Moisture	1/	1-0-1	3.3.2	4.5.3
	Disc (Dwg. 9366605) Exam. for Defects	15	15-0-1	3.6	Visual
	Foam (Dwg. 9366604) Workmanship	15	15-0-1	3.6	Visual
	Paper, Filter (Dwg. 9366607) Workmanship	15	15-0-1	3.6	Visual
	Wadding (Dwg. 9366606) Workmanship	15	15-0-1	3.6	Visual

NOTES:
1/ a sample of 10 gram shall be selected for inspection.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70668(AR)

PARAGRAPH	TITLE: Cartridge Assembly and Components		SHEET 2 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Case Assembly with Primer (Dwg. 9366635)	25			
	Workmanship		25-0-1	3.1	Visual
	Size		25-0-1	3.2	Gage
	Primer Sensitivity		25-0-1	3.4.1	4.5.4.1
	Pressure Cartridge Assembly (Dwg. 9366511)	150			
	Examination for Defects		150-0-1	3.2	4.4.2.2
	Primer Sensitivity 1/		150-0-1	3.4.1	4.5.4.1
	Functioning - Pressure/Time 1/				
	HOT		75-0-1	3.5.2	4.5.4.1
	COLD		75-0-1	3.5.3	4.5.4.2
	Propellant				
	M1, Form C Type I - (MIL-STD-652)				
	Grain Diameter	1/	1-0-1	3.1.1	Gage
	Moisture	1/	1-0-1	3.3.1	4.5.2

NOTES:
1/ a sample of 10 gram shall be used for inspection.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall compl with the lot formation Provisions of MIL-A-48078. In addition, inspection lots of cartridge assemblies shall contain:

a. M1 Propellant from not more than one lot from one manufacturer.

b. Propellant black powder from not more than one lot from one manufacturer.

c. Cartridge case assembly with primer from not more than one lot from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Single Sample Inspection Level II, Switching, By A.Q.L.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70668(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cartridge Case Assembly with Primer				9366635
					NEXT HIGHER ASSEMBLY 9366511
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Cartridge Gauge Improper		0.40%	3.2	Gage
102	Primer sensitivity Improper		0.40%	3.5.1	4.5.4.1
Minor					
201	Poor workmanship		1.00%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70668(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Black Powder				MIL-P-223
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Composition		Each lot	3.1.2	Certification
102	Moisture		.40	3.3.2	4.5.3
Minor	None defined				

NOTES:

AMSMC(m 1570, 1 Feb 85 Repl(. DRSMC-QA (D) Form 160, 1 Aug 83, which may not(used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70668(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Propellant				MIL-STD-652
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Grain diameter		.40	3.1.1	Gage
102	Composition		Each lot	3.1.1	Certification
103	Moisture			3.3.1	4.5.2
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70668(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Pressure Cartridge Assembly		9366511
			NEXT HIGHER ASSEMBLY: 9366491, 9385171

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Exposed Propellant Improper		100%	3.2	Visual
Major					
101	Outside diameter tube improper		0.40%	3.2	Gage
102	Diameter base improper		0.40%	3.2	Gage
103	Weight of black powder propellant improper		0.40%	3.2	Balance
104	Weight of M1 propellant improper		0.40%	3.2	Balance
105	Disk missing or incorrect		0.40%	3.2	Visual
106	Impact sensitivity, primer	1/	0.40%	3.5.1	4.5.5.1
107	Functioning - pressure/time		0.40%	3.5	4.5.5.2
Minor					
201	Overall length improper		0.65%	3.2	Gage
202	Primer joint not sealed with sealing compound		0.65%	3.2	Visual
203	Crimp missing or incorrect		0.65%	3.2	Visual
204	Crimp not sealed with sealing compound		0.65%	3.2	Visual
205	Poor workmanship		1.00%	3.7	Visual

NOTES:
1/ sample shall be equally divided for the hot and cold tests.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the “paragraph reference/inspection method” column of the First Article or Quality Conformance Inspection Tables herein. the contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirement by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Moisture content of M1 propellant. The moisture content of M1 propellant shall be determined at the time of loading. The moisture content shall be determined in accordance with MIL-STD-286, Method 101.3.2.

4.5.3 Moisture content of black powder propellant. The moisture content of black powder propellant shall be determined at the time of loading. The moisture content shall be determined in accordance with MIL-STD-286, Method 101.3.2.

4.5.4 Environmental conditions.

4.5.4.1 High temperature function. Units shall be stabilized per MIL-STD-810, Method 501.2 at +140°F +5°F and tested in accordance with 4.5.5.

4.5.4.2 Low temperature function. Units shall be stabilized per MIL-STD-810, Method 502.2 at -30°F -5°F and tested in accordance with 4.5.5.

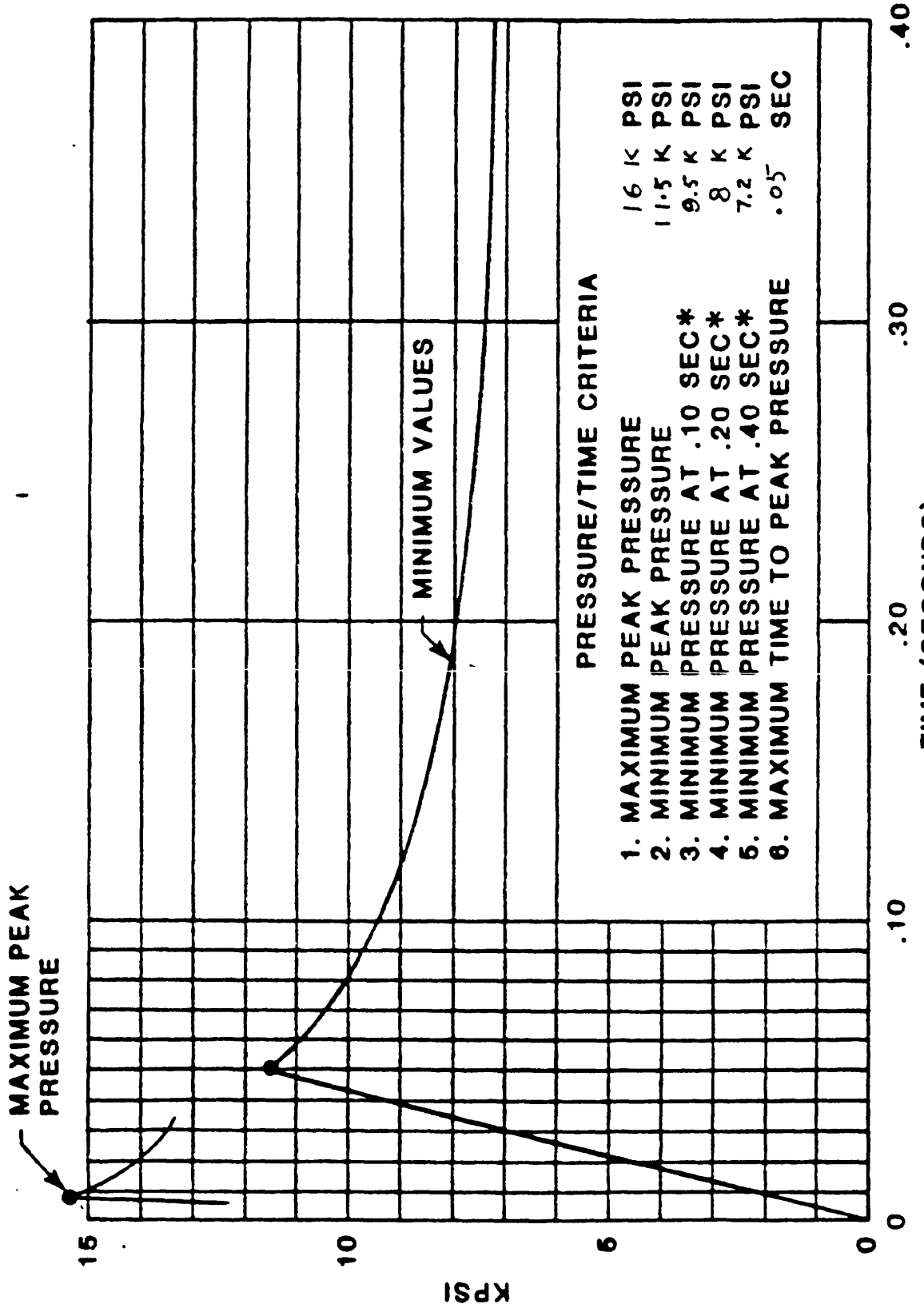
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4.5.5 Functioning.

4.5.5.1 Impact sensitivity primer. The sample units shall be mounted in a test fixture. The primer shall be impacted on centerline with a 1.94 ± .02 ounce firing pin with a tip of approximately .100 spherical diameter, impacting at 8.7 fps (14 inch drop) maximum. Statistical data shall be gathered to show 8.7 fps to be greater than the mean velocity plus 4 sigma deviations.

4.5.5.2 Pressure/time values. Sample cartridge assemblies, Dwg . 9366511 shall be selected in accordance with 4.4. The samples shall be assembled to a closed bomb type test fixture and the pressure/time values shown in Figure 1 shall be demonstrated by functioning the pressure cartridge.

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*ALL TIMES REFERENCED TO ZERO TIME SWITCH CLOSURE

FIGURE 1. PRESSURE/TIME CRITERIA FOR THE CARTRIDGE

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5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level C. The cartridge assemblies shall be prepared and packed in conformance with Department of Transportation Regulations, a part of the code of Federal Regulations, 49 CFR 173.94.

5.2 Packing.

5.2.1 Level C. The containers shall be packed in conformance with Department of Transportation Regulations, a part of the code of Federal Regulations, 49 CFR 173.94.

5.3 Markings. The containers shall be marked in conformance with MIL-STD-129.

6. NOTES

6.1 Ordering data. See MIL-A-48078 (AR).

6.2 Submission of inspection equipment designs for approval. (See MIL-A-58078). Submit equipment designs as required, to Commander: US ARDEC ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.3 Data submissions. Results of all tests performed by the contractor shall be submitted in accordance with Data Item DI-R-1721 on the DD Form 1423 for the contract. Distribution shall include the following:

Commander
ARDEC
ATTN : AMSMC-QAT-M (D)
Picatinny Arsenal, NJ 07806-5000

6.4 Distribution of ammunition data cards, inspection and test results. Distribution of ammunition data cards, inspection, and test results shall include Commander: ARDEC, Picatinny Arsenal, NJ 07806-5000, ATTN: AMSMC-QAT-M (D).

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research Development, and Engineering Center (ARDEC) may also include drawings prepared by and identified as, US Array Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.6 Subject term (key word) listing.
Pressure Cartridge Assembly
Cartridge Case Assembly

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A477)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70668	PRESSURE CARTRIDGE ASSEMBLY LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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